Agricultural Livelihoods and MSME Development Project (ALMDP)

Baseline Assessment Report

Kaduna State, Nigeria

Synthetic/Simulated Monitoring & Evaluation Portfolio Project

Baseline Year: 2026

Implementation Period: January 2027 - December 2030

Target Beneficiaries: 10,000

8,000 Smallholder Farmers | 2,000 Agricultural MSMEs

Important Note

This report is part of a synthetic/simulated Monitoring & Evaluation portfolio project. The dataset, beneficiary records, baseline results, and findings presented in this report are simulated for professional demonstration and analytical learning purposes and do not represent actual ALMDP implementation data or actual Kaduna State beneficiary records.

Prepared as an M&E Portfolio Demonstration

Reporting Focus: Baseline Assessment, Results Measurement, Data Quality, Indicator Analysis, and Management Use

1. Executive Summary

The Agricultural Livelihoods and MSME Development Project (ALMDP) is a four-year agricultural livelihoods and enterprise development initiative designed to support 10,000 beneficiaries in Kaduna State, comprising 8,000 smallholder farmers and 2,000 agricultural MSMEs. The project is designed to improve agricultural productivity, market participation, agricultural MSME performance, and the economic resilience and inclusion of women and young people.

This baseline assessment establishes the pre-implementation measurement position for the project's results framework. The assessment uses a synthetic cross-sectional dataset of 2,000 simulated respondents across seven project LGAs: Chikun, Igabi, Kachia, Kajuru, Kaduna North, Kaduna South, and Zaria. The dataset represents 1,600 farmers and 400 agricultural MSMEs across eight agricultural value chains.

Overall, the baseline shows a mixed performance position. Two indicators already meet or exceed their 2030 targets: average annual beneficiary income and adoption of at least three recommended agricultural practices. Agricultural productivity is also relatively close to its target. However, substantial gaps exist in economic resilience, market participation, buyer access, women's economic outcomes, post-harvest loss reduction, and agricultural MSME financial performance.

The baseline indicates that **30.60% of beneficiaries report improved economic resilience**, compared with a 2030 target of 65%. Only **27.31% of farmers sell through structured markets**, against a target of 55%, while **30.00% report improved buyer access**, compared with a target of 70%. Women's economic outcomes are also weak at **29.12%**, against a target of 60%.

Agricultural practice adoption presents a stronger starting position. **67.81% of farmers report adopting at least three recommended practices**, exceeding the 2030 target of 65%. Average crop productivity is **2.448 tonnes per hectare**, compared with a target of 2.50 tonnes per hectare. The official baseline post-harvest loss rate is **17.33%**, which remains above the 2030 target of 12%.

Among agricultural MSMEs, average annual profit is **NGN 356,839**, compared with a 2030 target of NGN 550,000. Half of assessed MSMEs maintain at least three basic financial record types, indicating a significant opportunity for business and financial management support.

The baseline also identifies important differences across agricultural value chains. Tomato has the highest average productivity at approximately **5.42 tonnes per hectare** and the highest average annual income at approximately **NGN 3.62 million**, while groundnut records the lowest average productivity at approximately **1.37 tonnes per hectare** and the lowest average annual income at approximately **NGN 876,347**.

Five indicators cannot be measured at baseline because the required implementation or monitoring data are not yet available. These include trained farmers demonstrating adequate knowledge, MSMEs reporting increased annual revenue, scheduled M&E activities completed on time, required reports submitted on time, and sampled records passing DQA.

Priority Management Areas

The baseline therefore establishes six major management priorities:

1. **Strengthen economic resilience** through climate-smart production, livelihood diversification, risk management, and targeted resilience interventions.
2. **Improve market participation and buyer access** through structured market linkages, market information, aggregation, and buyer engagement.
3. **Reduce post-harvest losses** through improved storage, handling, transportation, processing, and value-chain-specific interventions.
4. **Strengthen agricultural MSME profitability and financial management** through business development services, financial literacy, record keeping, and market support.
5. **Improve women's economic outcomes** by ensuring that participation translates into measurable income and livelihood gains.
6. **Establish implementation-period measurement systems** for indicators that cannot yet be assessed at baseline.

The baseline provides a reference point against which future monitoring data can be compared. It should therefore be used as a management tool for target setting, intervention prioritisation, performance monitoring, data quality assurance, and adaptive project management throughout implementation.

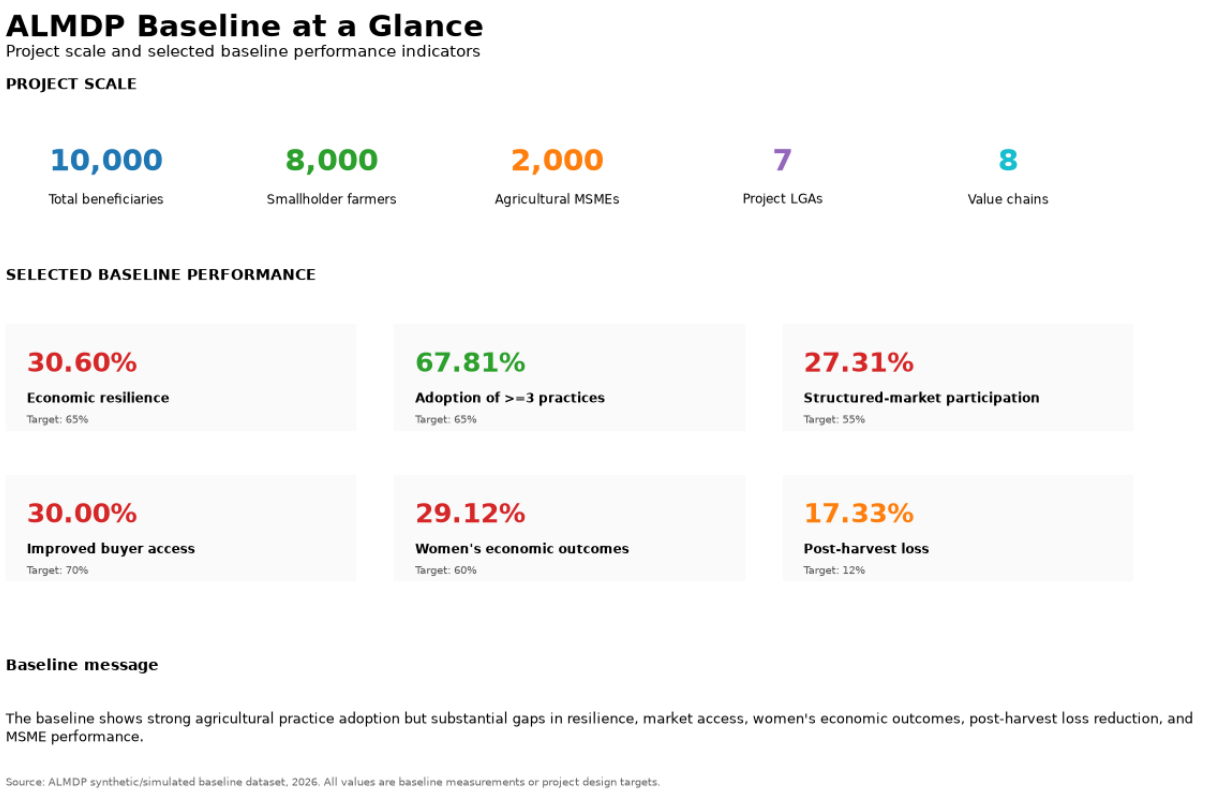


Figure 1. Executive summary: project scale and baseline performance.

2. Project Background

The Agricultural Livelihoods and MSME Development Project (ALMDP) is a four-year agricultural development initiative designed for Kaduna State, Nigeria. The project will be implemented from January 2027 to December 2030 and is designed to reach 10,000 direct beneficiaries, comprising 8,000 smallholder farmers and 2,000 agricultural MSMEs.

The project responds to livelihood and enterprise constraints affecting agricultural households and businesses, including low productivity, limited access to structured markets, post-harvest losses, weak business management systems, limited financial record keeping, and vulnerability to economic and climate-related shocks.

The project is designed around an integrated livelihoods and value-chain approach. Rather than focusing only on agricultural production, ALMDP combines farmer productivity interventions with market access, agricultural MSME development, inclusion, resilience, and institutional monitoring capacity.

Project Coverage

The synthetic project portfolio covers seven LGAs:

LGA	Baseline Respondents	Share
Chikun	350	17.50%
Kaduna North	300	15.00%
Kaduna South	300	15.00%
Igabi	300	15.00%
Kajuru	250	12.50%
Kachia	250	12.50%
Zaria	250	12.50%
Total	**2,000**	**100.00%**

The agricultural component covers eight value chains:

- Maize
- Rice
- Soybean
- Ginger
- Tomato
- Vegetables
- Sorghum
- Groundnut

Development Context

Smallholder farmers and agricultural MSMEs frequently face interconnected constraints. Production challenges can reduce yields and household income, while limited market access can reduce the prices received by producers. Post-harvest losses can further reduce the quantity and value of agricultural products available for sale. For agricultural MSMEs, weak financial management and limited business records can constrain profitability, planning, and access to finance.

ALMDP is therefore structured to address these constraints through complementary interventions. Farmer training and climate-smart practices are intended to strengthen production performance; market-linkage activities are intended to improve market participation; MSME support is intended to strengthen business performance; and inclusion and resilience activities are intended to ensure that project benefits extend to women, youth, and economically vulnerable beneficiaries.

Role of Monitoring and Evaluation

Monitoring and Evaluation is positioned as an integral management function within ALMDP. The M&E system is designed to establish a clear chain from project inputs and activities through outputs and outcomes to longer-term impact.

The baseline assessment provides the starting measurement point for this system. It establishes the initial values of measurable indicators, identifies indicators that require implementation-period data before they can be measured, highlights performance gaps against 2030 targets, and identifies priority areas for management attention.

Because this portfolio is synthetic, all beneficiary records, baseline measurements, results, and findings presented in this report are simulated. They are intended to demonstrate how an M&E professional could design, validate, analyse, interpret, and report baseline evidence within a structured development project environment.

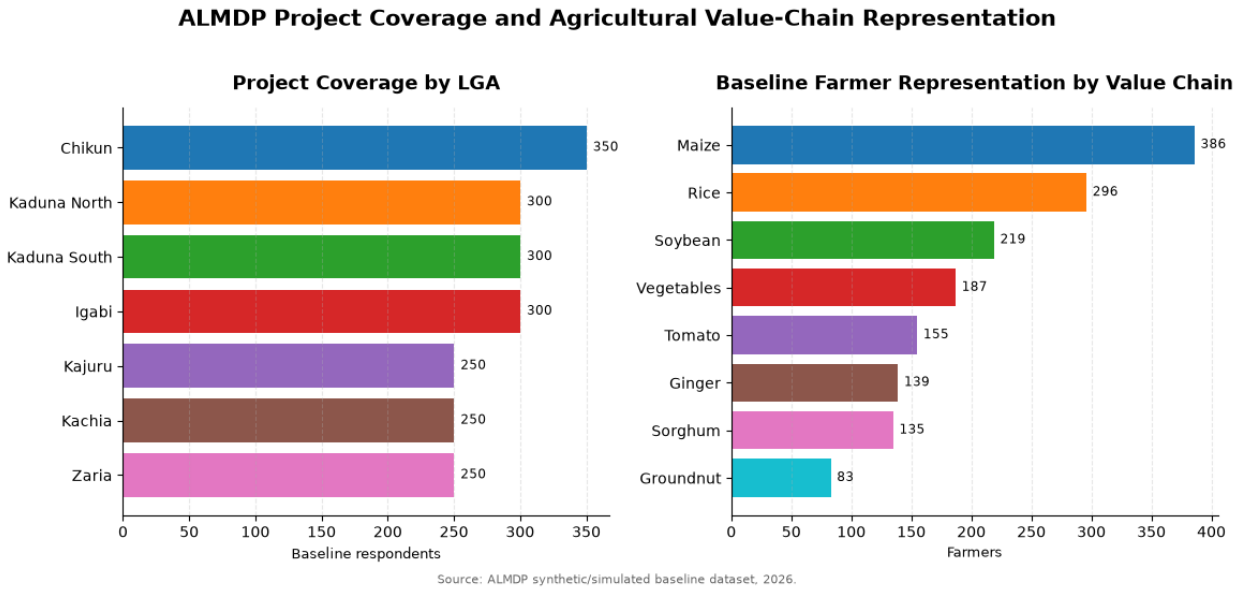


Figure 2. Project coverage and agricultural value-chain representation.

3. Project Goal and Objectives

3.1 Project Goal

The overall goal of the Agricultural Livelihoods and MSME Development Project is to **improve the productivity, income, market participation, and economic resilience of smallholder farmers and agricultural MSMEs in Kaduna State, Nigeria.**

The project is designed to achieve this through an integrated package of agricultural productivity support, market access and value-chain development, agricultural MSME development, and inclusion and institutional capacity strengthening.

3.2 Project Objectives

The project has four primary objectives:

Objective 1 - Improve Agricultural Productivity and Adoption

Increase the adoption of improved agricultural practices and technologies among smallholder farmers and improve agricultural productivity across the supported value chains.

Key areas include:

- Improved seed adoption
- Soil testing and improved soil management
- Fertilizer management
- Integrated pest management
- Conservation agriculture
- Water management
- Improved storage
- Climate-smart agricultural practices

Objective 2 - Improve Market Access and Participation

Improve farmers' ability to participate in structured markets and strengthen access to reliable buyers and relevant market information.

Key areas include:

- Structured market participation
- Buyer linkages
- Market information
- Market-linkage events
- Farmer aggregation
- Value-chain coordination

Objective 3 - Strengthen Agricultural MSME Performance

Improve the operational and financial performance of agricultural MSMEs through business development and financial management support.

Key areas include:

- Business management
- Financial record keeping
- Revenue and profitability improvement
- Financial management training
- Business planning
- Market development

Objective 4 - Strengthen Inclusion and Economic Resilience

Increase the participation and economic benefits of women and young people while strengthening beneficiaries' ability to withstand agricultural, economic, climate, health, and market-related shocks.

Key areas include:

- Women's participation
- Youth participation
- Women's income outcomes
- Climate and economic resilience
- Livelihood diversification
- Risk management

3.3 Project Components

The objectives will be implemented through four complementary project components:

Component	Primary Focus
Component 1: Farmer Productivity & Climate-Smart Agriculture	Agricultural productivity, improved practices, climate-smart agriculture, demonstrations and farmer training
Component 2: Market Access & Value-Chain Development	Market information, structured markets, buyer linkages, aggregation and value-chain coordination
Component 3: Agricultural MSME Development	Business support, financial management, record keeping, profitability and enterprise development
Component 4: Inclusion, Monitoring & Institutional Capacity	Gender, youth, resilience, M&E systems, data quality and institutional capacity

3.4 Intended Results

The project is ultimately intended to contribute to:

- Higher agricultural productivity.
- Increased adoption of recommended agricultural practices.
- Improved farmer knowledge and skills.
- Greater participation in structured markets.
- Improved access to agricultural buyers.
- Reduced post-harvest losses.
- Improved agricultural MSME revenue and profitability.
- Stronger financial management among MSMEs.
- Increased participation of women and youth.
- Improved economic resilience among beneficiaries.
- Stronger institutional monitoring and data-quality systems.

These objectives provide the basis for the project's results framework and the indicators assessed through the baseline.

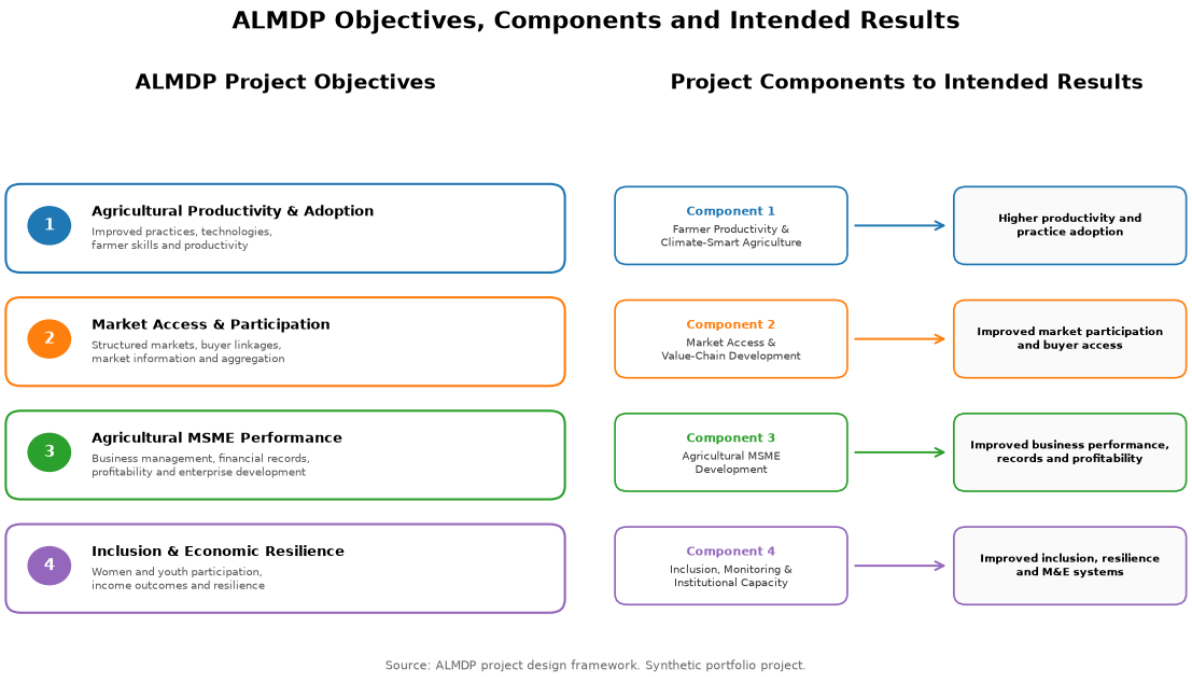


Figure 3. Project objectives and component-to-result architecture.

4. M&E Framework and Results Architecture

The ALMDP Monitoring and Evaluation framework is structured around a results-based management approach. It links project resources and activities to measurable outputs, outcomes, and longer-term impact. The framework is designed to support both accountability and management learning by providing evidence on whether implementation is producing the intended results.

4.1 Results Chain

The project's results architecture follows the sequence:

Inputs → Activities → Outputs → Short-term Outcomes → Medium-term Outcomes → Impact

Inputs

Inputs represent the resources required to implement the project, including:

- Financial resources
- Technical expertise
- Project personnel
- Training resources
- Agricultural technologies and materials
- Monitoring and data-management systems
- Institutional partnerships

Activities

Activities represent the interventions delivered by the project, including:

- Farmer training
- Climate-smart agriculture training
- Demonstration plot establishment
- Market-linkage activities
- Market information dissemination
- Agricultural MSME business support
- Financial management training
- Gender and youth inclusion activities
- Monitoring, data collection and data-quality activities

Outputs

Outputs represent the immediate products and services delivered by the project.

The output-level indicators include:

- **OP-01:** Number of farmers trained
- **OP-02:** Number of farmers completing CSA training

- **OP-03:** Number of demonstration plots established
- **OP-04:** Number of market-linkage events
- **OP-05:** Number of farmers receiving market information
- **OP-06:** Number of MSMEs receiving business support
- **OP-07:** Number of MSMEs completing financial management training
- **OP-08:** Percentage of scheduled M&E activities completed on time
- **OP-09:** Percentage of required reports submitted on time
- **OP-10:** Percentage of sampled records passing DQA

Outcomes

Outcome indicators measure changes in beneficiary behaviour, performance, access, and economic conditions.

The agricultural outcome indicators include:

- **OUT-01:** Average crop productivity
- **OUT-02:** Farmers adopting at least three recommended practices
- **OUT-03:** Trained farmers demonstrating adequate knowledge
- **OUT-04:** Farmers selling through structured markets
- **OUT-05:** Farmers reporting improved buyer access
- **OUT-06:** Average post-harvest loss rate

Agricultural MSME outcome indicators include:

- **OUT-07:** Supported MSMEs reporting increased annual revenue
- **OUT-08:** Average MSME annual profit
- **OUT-09:** MSMEs maintaining basic financial records

Inclusion and economic outcome indicators include:

- **OUT-10:** Direct beneficiaries who are women
- **OUT-11:** Direct beneficiaries aged 18-35
- **OUT-12:** Women beneficiaries reporting increased agricultural or business income

Impact

The impact level captures longer-term changes in beneficiary economic well-being and resilience:

- **IMP-01:** Average annual beneficiary income
- **IMP-02:** Beneficiaries reporting improved economic resilience

4.2 M&E Data Flow

The ALMDP M&E system follows a controlled data flow:

Field Data → Data Collection → Field Review → Submission → Automated Validation → Manual QA → Clean Dataset → Indicator Calculation → M&E Database → Dashboard/Report → Management Decision

This process is designed to ensure that information used for reporting is traceable from the original source through validation and analysis to the final management product.

4.3 Data Quality Assurance

Data quality is assessed across five core dimensions:

Dimension	M&E Focus
Accuracy	Whether recorded values correctly represent the underlying information
Completeness	Whether required records and fields are available
Consistency	Whether values and relationships remain logically consistent
Timeliness	Whether data are submitted and processed within required timelines
Integrity	Whether data remain protected from inappropriate alteration or manipulation

The baseline workflow applies these principles through data profiling, validation rules, duplicate checks, logical consistency checks, derived-variable recalculation, cleaning logs, validation logs, and post-cleaning verification.

4.4 Verification and Data Use

The framework incorporates multiple verification mechanisms, including:

- Source document review
- Beneficiary verification
- Attendance-register review
- Digital validation checks
- Spot checks
- Back-checks
- Record-to-report reconciliation
- Duplicate detection
- Calculation verification
- Periodic Data Quality Assessments

The purpose of verification is not only to identify errors but also to strengthen confidence in the evidence used for programme decisions.

4.5 Results-Based Management

The M&E system is designed to support a continuous management cycle:

Identify Variance → Investigate Cause → Document Evidence → Define Corrective Action → Assign Responsibility → Set Deadline → Monitor Follow-up

This approach allows project management to use monitoring evidence to identify implementation bottlenecks, respond to underperformance, document corrective measures, and track whether actions produce improvement.

4.6 Baseline Measurement Role

The baseline assessment provides the reference point against which subsequent monitoring and evaluation results will be compared.

For each measurable indicator, the baseline establishes:

- Initial performance level
- Valid numerator and denominator where applicable
- Measurement unit
- Direction of desired change
- Relevant disaggregation
- Data availability
- Gap relative to the 2030 target

Indicators that cannot be measured at baseline are explicitly identified rather than assigned artificial values. This distinction is important for maintaining the integrity of the results framework and ensuring that future measurements are comparable with the baseline.

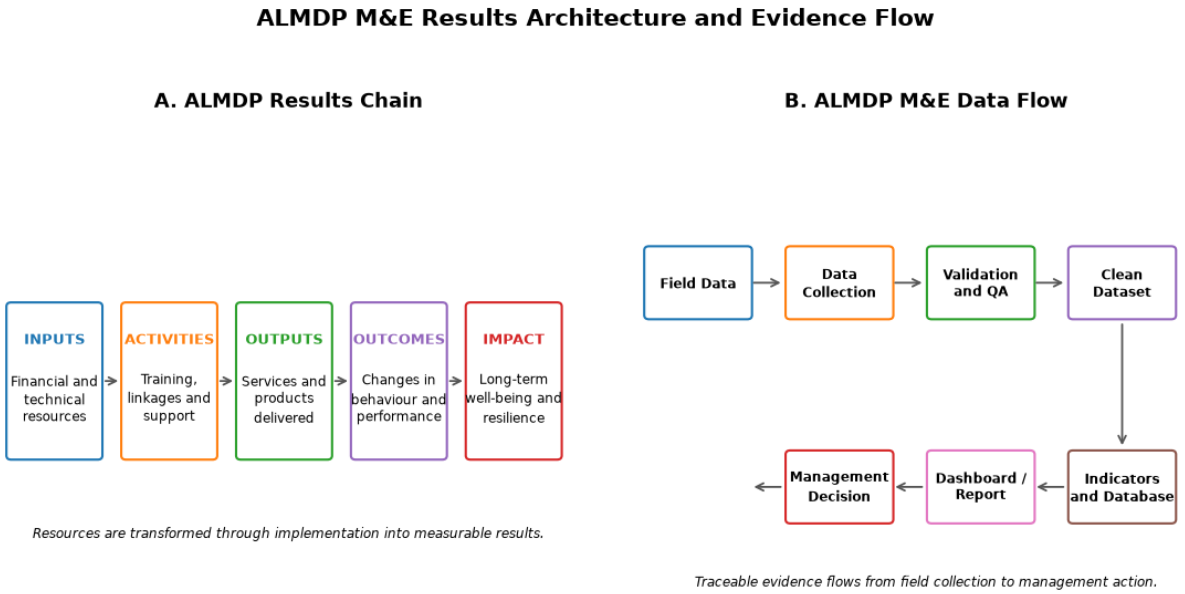


Figure 4. Results architecture and M&E; data flow.

5. Baseline Methodology

5.1 Assessment Design

The baseline assessment uses a **synthetic cross-sectional assessment design** developed to demonstrate a complete Monitoring and Evaluation workflow for the ALMDP portfolio.

The assessment represents the measurement position immediately before project implementation. The baseline is therefore treated as a pre-implementation reference point against which future project performance can be assessed.

The analysis uses a structured synthetic beneficiary dataset containing 2,000 simulated respondent records. The dataset represents two beneficiary groups:

- 1,600 smallholder farmers
- 400 agricultural MSMEs

The dataset covers seven project LGAs and eight agricultural value chains.

Synthetic Data Disclosure: *All respondent records, measurements, baseline values, and findings in this report are simulated for professional portfolio demonstration and learning purposes. They do not represent actual ALMDP beneficiaries, actual project implementation records, or official Kaduna State statistics.*

5.2 Baseline Data Source

The primary analytical dataset is a structured synthetic baseline dataset generated to reflect the types of information that would normally be collected through a beneficiary baseline assessment.

The dataset contains information relating to:

- Respondent identification and survey administration
- Geographic location
- Beneficiary type
- Sex and age
- Education
- Household characteristics
- Agricultural production
- Agricultural practices
- Market participation
- Market information
- Post-harvest losses
- Agricultural and non-agricultural income
- Agricultural MSME characteristics
- Business performance
- Financial record keeping
- Economic resilience
- Knowledge and practice-related variables

5.3 Sampling and Respondent Coverage

The synthetic baseline includes **2,000 respondents** distributed across the seven project LGAs.

The respondent structure is:

Category	Number	Share
Smallholder farmers	1,600	80.00%
Agricultural MSMEs	400	20.00%
Total	**2,000**	**100.00%**

The farmer sample covers eight value chains:

- Maize
- Rice
- Soybean
- Ginger

- Tomato
- Vegetables
- Sorghum
- Groundnut

5.4 Data Collection Instrument

The baseline is based on a structured beneficiary questionnaire designed around the project's results framework and indicator requirements.

The questionnaire domains include:

1. Respondent identification and consent
2. Demographic characteristics
3. Household and socioeconomic characteristics
4. Agricultural production and productivity
5. Agricultural practice adoption
6. Market access and participation
7. Post-harvest losses
8. Income and livelihood sources
9. Agricultural MSME characteristics
10. Business performance
11. Financial management
12. Gender and youth inclusion
13. Economic resilience
14. Knowledge and training-related variables

The questionnaire structure was designed to ensure that baseline data requirements could be traced to the indicator register and results framework.

5.5 Data Processing and Analysis

The baseline data-processing workflow followed the sequence:

Raw Data → Profiling → Validation → Cleaning → Derived Variable Recalculation → Post-Cleaning QA → Indicator Calculation → Disaggregation → Analysis → Reporting

The raw dataset was first profiled to identify missing values, duplicate identifiers, invalid responses, logical inconsistencies, extreme values, and other potential data-quality issues.

Validation rules were then applied to identify records requiring correction or review.

Cleaning procedures included:

- Resolution of respondent ID collisions
- Conversion of invalid age values to missing
- Treatment of non-positive farm-area values as missing
- Correction of logically impossible production relationships
- Removal of invalid knowledge responses
- Recalculation of derived variables
- Preservation of valid extreme observations for review

The cleaned dataset retained the same analytical population and variable structure as the raw dataset.

5.6 Indicator Calculation

Indicators were calculated using their approved definitions and measurement rules.

For percentage indicators, the calculation generally followed:

Indicator (%) = Valid Numerator / Valid Denominator × 100

For continuous indicators, valid observations were used to calculate the relevant mean or aggregate measure according to the indicator definition.

Where an indicator required a lower value to represent better performance, the direction was explicitly retained. This applies particularly to **OUT-06: Average post-harvest loss rate**.

Indicator calculations were based on valid observations rather than forcing missing or unavailable values into the analysis.

5.7 Disaggregation

Baseline results were disaggregated where relevant to support equity analysis and management decision-making.

Key disaggregation dimensions included:

- LGA
- Beneficiary type
- Sex

- Age/youth status
- Agricultural value chain

Disaggregation was applied particularly to indicators relating to inclusion, resilience, agricultural productivity, market participation, and economic performance.

5.8 Treatment of Missing and Unavailable Indicators

Missing observations were not automatically interpreted as zero.

Where individual observations were missing, the indicator denominator was based on valid observations where consistent with the approved indicator definition.

Where an entire indicator could not be calculated because the required implementation-period variable was unavailable, the indicator was classified as **Not measurable** rather than assigning an artificial baseline value.

At baseline, the following indicators were not measurable:

- **OUT-03:** Trained farmers demonstrating adequate knowledge
- **OUT-07:** Supported MSMEs reporting increased annual revenue
- **OP-08:** Scheduled M&E activities completed on time
- **OP-09:** Required reports submitted on time
- **OP-10:** Sampled records passing DQA

This approach preserves the distinction between zero performance, missing observations, and indicators that appropriately require implementation-period data.

5.9 Outlier Treatment

Extreme observations were reviewed as part of the data-quality process.

Valid but unusually large observations were retained where they did not violate the defined logical or validity rules. This approach avoids unnecessarily removing genuine variation from the synthetic population.

Outlier review therefore focused on determining whether an observation was:

- Logically invalid
- Outside an allowable range
- Inconsistent with related variables
- Unusually large but potentially valid

Only observations failing the relevant validity or logical checks were corrected or treated as missing.

5.10 Analytical Limitations

Because the assessment is synthetic, the findings should not be interpreted as estimates of actual Kaduna State agricultural conditions or actual ALMDP beneficiary performance.

The methodology is intended to demonstrate:

- M&E system design
- Baseline indicator calculation
- Data-quality management
- Disaggregation
- Performance-gap analysis
- Evidence-based interpretation
- Management-oriented reporting

The analytical approach nevertheless follows the structure and quality-control principles expected in a professional baseline assessment.

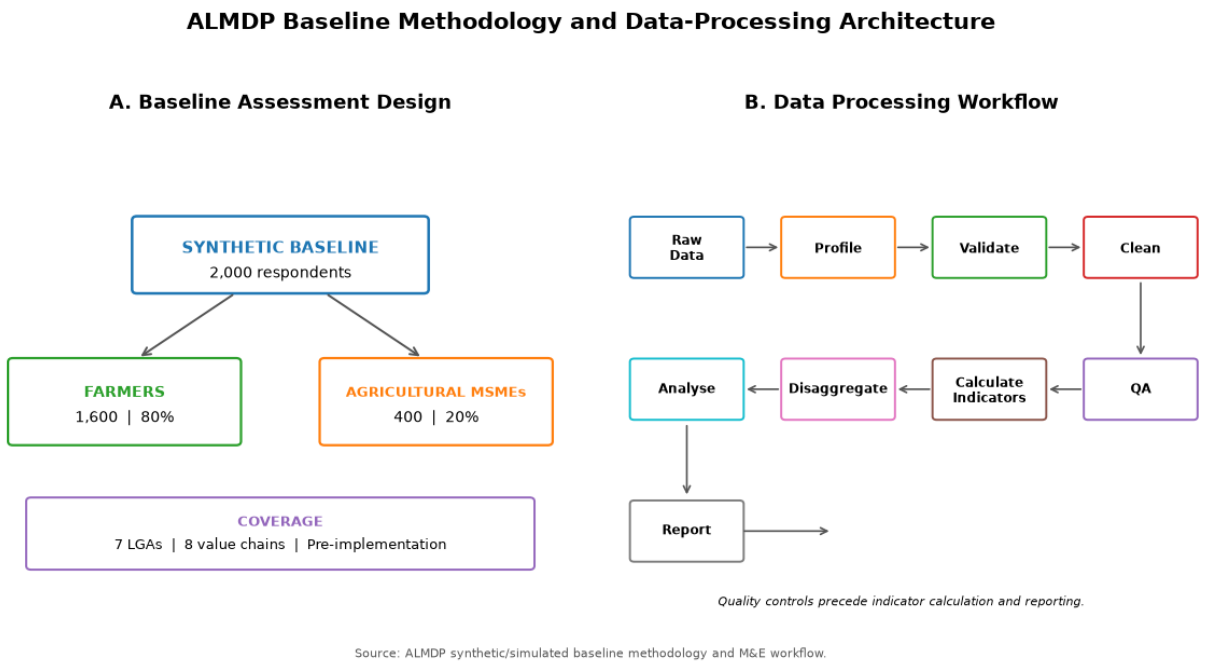


Figure 5. Baseline design and data-processing workflow.

6. Sampling and Respondent Profile

6.1 Respondent Coverage

The synthetic baseline dataset contains **2,000 simulated respondents** representing the two principal beneficiary groups targeted by ALMDP.

Beneficiary Group	Respondents	Share
Smallholder farmers	1,600	80.00%
Agricultural MSMEs	400	20.00%
Total	**2,000**	**100.00%**

The respondent structure provides analytical coverage of both agricultural production and agricultural enterprise activities within the project design.

6.2 Sex Distribution

Of the 2,000 respondents:

Sex	Respondents	Share
Female	831	41.55%
Male	1,169	58.45%
Total	**2,000**	**100.00%**

Women therefore represent **41.55% of direct beneficiaries** in the synthetic baseline population. This provides a baseline position for monitoring progress toward the project's 2030 target of 45%.

6.3 Age Profile

Age was available for **1,999 of the 2,000 respondents**.

Age Group	Respondents	Share of Valid Ages
18-35 years	603	30.17%
36-59 years	892	44.62%
60 years and above	504	25.21%
Total valid age observations	**1,999**	**100.00%**

The average age among respondents with valid age data was approximately **46.2 years**.

The youth category, defined as respondents aged 18-35, accounts for **30.17% of respondents with valid age information**. This provides the baseline position for OUT-11, which has a 2030 target of 35%.

6.4 Education Profile

The education distribution was:

Education Level	Respondents	Share
No formal education	224	11.20%
Primary	511	25.55%
Secondary	876	43.80%
Post-secondary	389	19.45%
Total	**2,000**	**100.00%**

Secondary education represents the largest category at **43.80%**, followed by primary education at **25.55%** and post-secondary education at **19.45%**.

This profile is relevant to the design of training, extension, business-development, and communication approaches. It suggests the importance of practical, accessible, and appropriately structured learning materials.

6.5 Geographic Distribution

Respondents were distributed across seven project LGAs:

LGA	Respondents	Share
Chikun	350	17.50%
Kaduna North	300	15.00%
Kaduna South	300	15.00%
Igabi	300	15.00%
Kajuru	250	12.50%
Kachia	250	12.50%
Zaria	250	12.50%
Total	**2,000**	**100.00%**

The largest respondent group is from Chikun, representing **17.50%** of the synthetic baseline population. Kaduna North, Kaduna South, and Igabi each contribute 15.00%, while Kajuru, Kachia, and Zaria each contribute 12.50%.

6.6 Farmer Value-Chain Distribution

The 1,600 farmer respondents are distributed across eight agricultural value chains:

Value Chain	Farmers	Share
Maize	386	24.13%
Rice	296	18.50%
Soybean	219	13.69%
Vegetables	187	11.69%
Tomato	155	9.69%
Ginger	139	8.69%
Sorghum	135	8.44%
Groundnut	83	5.19%
Total	**1,600**	**100.00%**

Maize is the largest farmer value chain, representing **24.13%** of farmers, followed by rice at **18.50%** and soybean at **13.69%**.

Groundnut represents the smallest farmer group at **5.19%**.

6.7 Respondent Profile Implications

The respondent profile provides several implications for implementation and monitoring:

- The project will need to maintain strong farmer-focused interventions because farmers represent 80% of the baseline population.
- Women's participation is already relatively close to the 2030 target, but participation must translate into measurable economic gains.
- Youth represent approximately 30% of respondents with valid age information, creating a substantial base for youth-focused interventions.
- The education profile supports the use of practical and accessible training approaches.
- The value-chain distribution suggests that maize, rice, and soybean interventions will reach a large share of farmers, while smaller value chains may require more targeted approaches.
- Geographic disaggregation will be important for identifying differences in implementation performance and beneficiary outcomes across LGAs.

The respondent profile therefore establishes the demographic, geographic, and sectoral context needed to interpret subsequent baseline indicator findings.

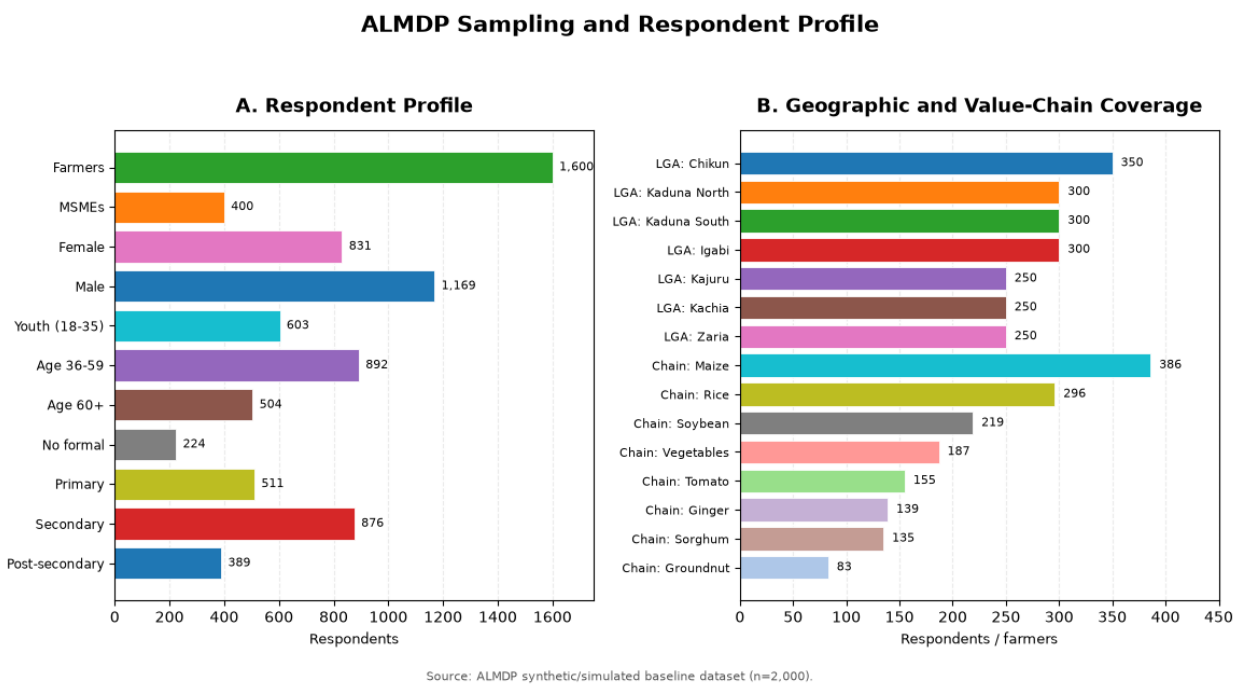


Figure 6. Respondent profile and geographic/value-chain coverage.

7. Data Quality Assurance and Cleaning

7.1 Data Quality Approach

Data Quality Assurance (DQA) was integrated into the baseline workflow to ensure that the dataset used for indicator calculation was sufficiently accurate, complete, consistent, timely, and reliable for analytical purposes.

The baseline DQA process followed the sequence:

Data Profiling → Validation → Issue Classification → Cleaning → Derived Variable Recalculation → Post-Cleaning Verification

The process distinguished between data-quality errors requiring correction and unusual observations that required review but were not necessarily invalid.

7.2 Data Quality Dimensions

Five core dimensions were applied:

DQA Dimension	Application
Accuracy	Verification that recorded values and derived indicators were correctly represented
Completeness	Assessment of missing required observations and fields
Consistency	Identification of contradictory or logically incompatible values
Timeliness	Assessment of whether data would be available within required reporting timelines
Integrity	Protection of the analytical dataset from inappropriate alteration and preservation of an auditable cleaning process

7.3 Raw Dataset Quality Issues

The synthetic raw dataset contained intentionally embedded quality issues to demonstrate a realistic M&E data-quality workflow.

The raw dataset contained:

- **2,000 records**
- **109 variables**
- **5 duplicate respondent-ID collisions**
- Missing household-size observations
- Missing crop-income observations
- Missing farm-area observations
- Invalid age values
- A negative farm-area value
- Invalid knowledge-response coding
- Production relationships requiring logical checks
- Extreme but reviewable values for farm size, business income, and production

These issues were identified through automated validation rules and subsequent review.

7.4 Validation

Automated validation was used to identify potential data-quality problems before cleaning.

Validation checks included:

- Duplicate respondent IDs
- Missing required fields
- Age-range validation
- Farm-area validation

- Production-volume validation
- Harvested-versus-sold quantity checks
- Harvested-versus-lost quantity checks
- Knowledge-response validation
- Derived-variable consistency checks
- Indicator calculation checks

The validation process classified identified issues according to their severity and whether they required correction, review, or further investigation.

7.5 Cleaning Procedures

The cleaning process retained the full analytical population while correcting invalid or logically inconsistent observations.

The principal cleaning actions were:

Issue	Cleaning Action
Duplicate respondent-ID collisions	IDs reassigned to maintain unique respondent identifiers
Invalid age values	Converted to missing
Non-positive farm area	Converted to missing
Quantity sold greater than quantity harvested	Invalid quantity treated as missing
Quantity lost greater than quantity harvested	Invalid quantity treated as missing
Invalid knowledge response	Converted to missing
Derived-variable inconsistencies	Derived variables recalculated
Extreme but potentially valid observations	Retained for review

No respondents were unnecessarily removed solely because their values were unusually high or low.

7.6 Post-Cleaning Verification

Following cleaning, the dataset was revalidated.

Key post-cleaning checks confirmed:

Quality Check	Result
Duplicate respondent IDs	0
Invalid age values	0
Non-positive farmer farm areas	0
Sold quantity greater than harvested quantity	0
Lost quantity greater than harvested quantity	0
Invalid knowledge responses	0
Invalid productivity calculations	0

The cleaned dataset retained **2,000 records and the full 109-variable structure**.

7.7 Derived Variable Quality Assurance

Derived variables were independently checked to ensure that calculations were internally consistent.

The derived-variable QA process produced:

- **8 derived-variable mismatch checks**
- **0 mismatches identified**

This provided an additional verification layer before the calculation of baseline indicators.

7.8 Indicator-Level Quality Assurance

Indicator QA was conducted after the cleaned dataset had been validated.

The indicator register was checked to confirm:

- All **24 indicators** were represented exactly once.
- Indicator IDs were unique.
- Numerators and denominators were available where required.
- Percentage indicators used valid denominators.
- Direction of measurement was correctly recorded.
- OUT-06 was correctly treated as a lower-is-better indicator.
- Indicators without required baseline data were explicitly classified as not measurable.
- Baseline results were traceable to the underlying dataset.

The indicator-level QA process passed.

7.9 Data Cleaning Documentation

The cleaning process was documented through dedicated project records, including:

- Validation rules
- Validation log
- Validation summary
- Cleaning log
- Raw dataset
- Clean dataset

This documentation creates an audit trail between the original synthetic data and the final dataset used for analysis.

7.10 Data Quality Management Implications

The baseline DQA exercise demonstrates the importance of establishing quality controls before reporting results.

For implementation, the project should maintain:

1. Standardized data-collection instruments.
2. Clear variable definitions and coding schemes.
3. Automated validation rules.
4. Field-level review before submission.
5. Supervisor-level quality checks.
6. Routine duplicate and consistency checks.
7. Documented cleaning procedures.
8. Version-controlled datasets.
9. Indicator calculation verification.
10. Periodic formal Data Quality Assessments.

These controls will reduce the risk of inaccurate reporting and strengthen confidence in evidence used for management decisions.

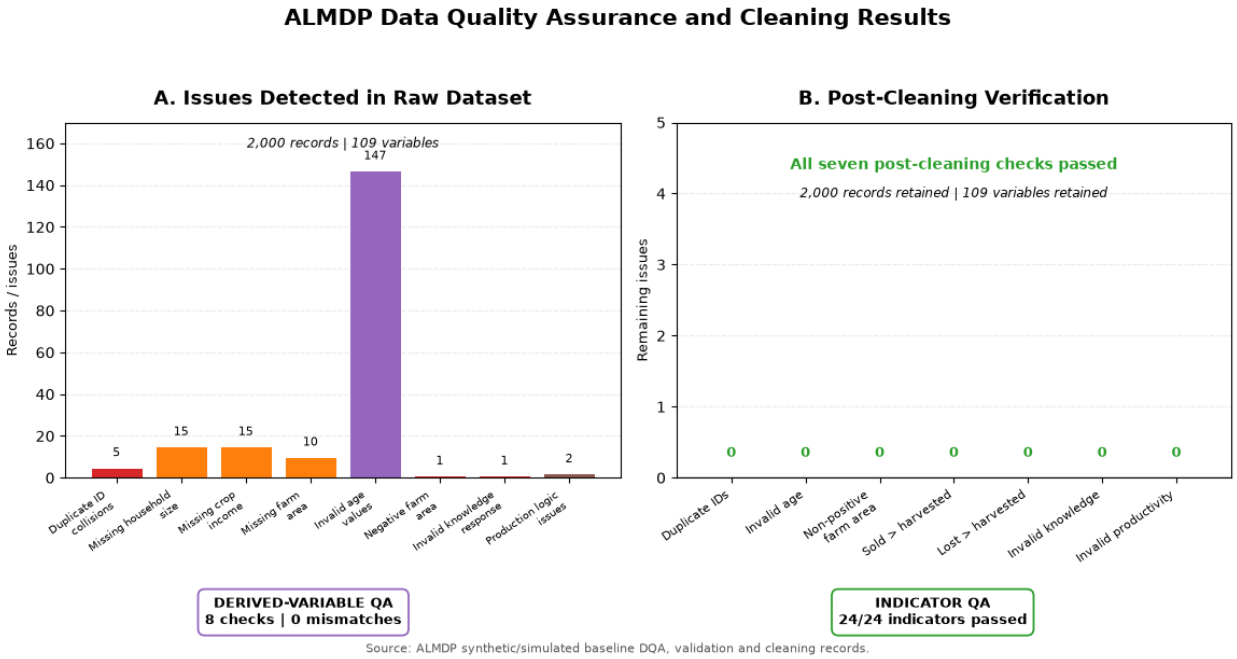


Figure 7. Data-quality issues before cleaning and validation status after cleaning.

8. Baseline Indicator Results

8.1 Overview

The baseline assessment measured 24 indicators across impact, outcome, and output levels of the ALMDP results framework.

The results provide the initial measurement position against which implementation progress will be assessed through 2030.

Of the 24 indicators:

- **12 outcome and impact indicators** were measurable from the baseline dataset.
- **2 additional outcome indicators** require implementation-period data and were therefore not measurable at baseline.
- **7 output indicators** are pre-implementation measures with a baseline value of zero.
- **3 M&E system-performance indicators** require implementation-period monitoring data and were not measurable at baseline.

The baseline results are presented below.

8.2 Impact Indicators

Indicator	Baseline	Unit	2030 Target	Direction	Status
IMP-01: Average annual beneficiary income	1,497,879.70	NGN/year	600,000	Higher is better	Target met/exceeded
IMP-02: Beneficiaries reporting improved economic resilience	30.60%	%	65%	Higher is better	Requires improvement

IMP-01 - Average Annual Beneficiary Income

The baseline average annual beneficiary income is **NGN 1,497,879.70**, substantially above the 2030 target of NGN 600,000.

This indicator therefore meets or exceeds the target at baseline.

The result should nevertheless be interpreted cautiously because the project is pre-implementation and the dataset is synthetic. The baseline should be retained as the formal reference value for future monitoring rather than assuming that the observed level represents project-generated performance.

IMP-02 - Economic Resilience

Only **30.60% of beneficiaries** report improved economic resilience at baseline, compared with a 2030 target of 65%.

This represents a **34.40 percentage-point gap** to the target and identifies economic resilience as a major programme priority.

8.3 Agricultural Productivity and Practice Adoption

Indicator	Baseline	Unit	2030 Target	Direction	Status
OUT-01: Average crop productivity	2.448	t/ha	2.50	Higher is better	Requires improvement
OUT-02: Farmers adopting at least three recommended practices	67.81%	%	65%	Higher is better	Target met/exceeded
OUT-03: Trained farmers demonstrating adequate knowledge	Not measurable	%	75%	Higher is better	Not measurable

OUT-01 - Average Crop Productivity

Average crop productivity is **2.448 tonnes per hectare**, compared with a 2030 target of 2.50 tonnes per hectare.

The baseline is therefore approximately **0.052 tonnes per hectare below the target**, indicating that the project begins relatively close to its intended productivity level.

OUT-02 - Adoption of Recommended Practices

67.81% of farmers report adopting at least three recommended agricultural practices.

This exceeds the 2030 target of 65%, indicating a comparatively strong starting position for agricultural practice adoption.

OUT-03 - Adequate Knowledge

OUT-03 cannot be measured at baseline because the required implementation-related training variable is unavailable.

It is therefore classified as **Not measurable** rather than assigning an artificial baseline value.

Future measurement should establish the proportion of trained farmers demonstrating adequate knowledge using the approved knowledge assessment and denominator definition.

8.4 Market Access and Post-Harvest Outcomes

Indicator	Baseline	Unit	2030 Target	Direction	Status
OUT-04: Farmers selling through structured markets	27.31%	%	55%	Higher is better	Requires improvement
OUT-05: Farmers reporting improved buyer access	30.00%	%	70%	Higher is better	Requires improvement
OUT-06: Average post-harvest loss rate	17.33%	%	12%	Lower is better	Requires improvement

OUT-04 - Structured Market Participation

Only **27.31% of farmers** report selling through structured markets at baseline, compared with a target of 55%.

The baseline is therefore **27.69 percentage points below the target**, indicating a substantial market-participation gap.

OUT-05 - Improved Buyer Access

30.00% of farmers report improved buyer access at baseline, compared with a target of 70%.

The resulting **40 percentage-point gap** makes buyer access one of the most significant outcome-level priorities.

OUT-06 - Post-Harvest Loss Rate

The official baseline post-harvest loss rate is **17.33%**, compared with a 2030 target of 12%.

Because lower performance values represent better performance for this indicator, the baseline is **5.33 percentage points above the target**.

The indicator should therefore be monitored using its lower-is-better direction rather than the standard higher-is-better interpretation used for most outcome indicators.

8.5 Agricultural MSME Outcomes

Indicator	Baseline	Unit	2030 Target	Direction	Status
OUT-07: MSMEs reporting increased annual revenue	Not measurable	%	60%	Higher is better	Not measurable
OUT-08: Average MSME annual profit	356,839.05	NGN/year	550,000	Higher is better	Requires improvement
OUT-09: MSMEs maintaining basic financial records	50.00%	%	70%	Higher is better	Requires improvement

OUT-07 - Increased MSME Revenue

OUT-07 cannot be measured at baseline because the required implementation-period revenue-change variable is unavailable.

The indicator will become measurable once project-supported MSMEs have implementation-period data that can be compared against their approved baseline and measurement definition.

OUT-08 - Average MSME Annual Profit

Average annual MSME profit is **NGN 356,839.05**, compared with a target of NGN 550,000.

This indicates a baseline shortfall of approximately **NGN 193,160.95** relative to the target.

OUT-09 - Basic Financial Records

50.00% of assessed MSMEs maintain at least three basic financial record types, compared with a target of 70%.

This indicates a **20 percentage-point gap** and demonstrates a clear opportunity for financial-management and business-development support.

8.6 Inclusion Outcomes

Indicator	Baseline	Unit	2030 Target	Direction	Status
OUT-10: Direct beneficiaries who are women	41.55%	%	45%	Higher is better	Requires improvement
OUT-11: Direct beneficiaries aged 18-35	30.17%	%	35%	Higher is better	Requires improvement
OUT-12: Women beneficiaries reporting increased agricultural/business income	29.12%	%	60%	Higher is better	Requires improvement

OUT-10 - Women Beneficiary Participation

Women represent **41.55% of direct beneficiaries**, compared with a target of 45%.

The baseline is relatively close to the target, with a gap of **3.45 percentage points**.

OUT-11 - Youth Participation

Respondents aged 18-35 represent **30.17% of beneficiaries with valid age information**, compared with a target of 35%.

The baseline gap is approximately **4.83 percentage points**.

OUT-12 - Women's Economic Outcomes

Only **29.12% of women beneficiaries** report increased agricultural or business income at baseline, compared with a target of 60%.

This represents a substantial **30.88 percentage-point gap** and indicates that participation alone will not be sufficient. The project will need to monitor whether women's participation translates into measurable economic gains.

8.7 Output Indicators

Output indicators are primarily pre-implementation measures. The baseline value is therefore zero for activities that have not yet been delivered.

Indicator	Baseline	Unit	2030 Target	Status
OP-01: Farmers trained	0	Farmers	8,000	Requires improvement
OP-02: Farmers completing CSA training	0	Farmers	5,000	Requires improvement
OP-03: Demonstration plots established	0	Plots	80	Requires improvement
OP-04: Market-linkage events	0	Events	40	Requires improvement
OP-05: Farmers receiving market information	0	Farmers	6,000	Requires improvement
OP-06: MSMEs receiving business support	0	MSMEs	2,000	Requires improvement
OP-07: MSMEs completing financial-management training	0	MSMEs	1,500	Requires improvement

These zero baselines should not be interpreted as negative performance. They represent the expected pre-implementation state before project activities begin.

8.8 M&E System Performance Indicators

Three indicators are not measurable at baseline because they depend on implementation and reporting processes that have not yet begun.

Indicator	Baseline Status	2030 Target
OP-08: Scheduled M&E activities completed on time	Not measurable	≥90%
OP-09: Required reports submitted on time	Not measurable	100%
OP-10: Sampled records passing DQA	Not measurable	≥95%

These indicators should be activated once implementation-period monitoring systems are operational.

8.9 Overall Baseline Interpretation

The baseline presents a mixed performance position.

Stronger Starting Positions

- Average annual beneficiary income exceeds the 2030 target.
- Adoption of at least three recommended agricultural practices exceeds the 2030 target.
- Average crop productivity is close to the target.
- Women's participation is relatively close to the target.
- Youth participation is also relatively close to the target.

Priority Gaps

- Economic resilience is substantially below target.
- Structured market participation is weak.
- Buyer access is substantially below target.
- Post-harvest losses remain above the desired level.
- MSME profitability is below target.
- Financial record keeping among MSMEs remains incomplete.
- Women's economic outcomes are substantially below target.

Not Yet Measurable

Five indicators require implementation-period information before they can be assessed:

- OUT-03
- OUT-07
- OP-08
- OP-09
- OP-10

The results establish the quantitative baseline against which subsequent implementation monitoring should be compared.

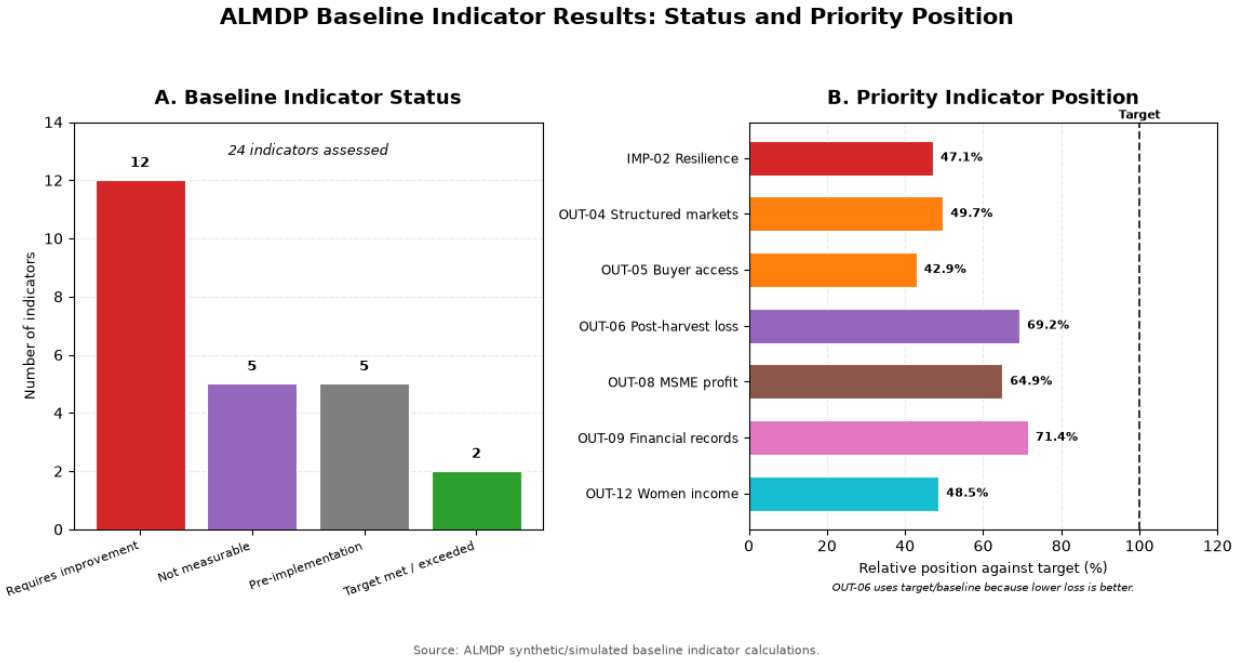


Figure 8. Baseline indicator status and priority gaps relative to 2030 targets.

9. Disaggregated Baseline Findings

Disaggregation provides a more detailed understanding of baseline conditions and helps identify where programme interventions may require differentiated approaches. The baseline was disaggregated by beneficiary type, LGA,

agricultural value chain, sex, age group, and selected thematic dimensions.

9.1 Beneficiary Type

The baseline population consists of 1,600 farmers and 400 agricultural MSMEs.

Measure	Farmers	MSMEs
Respondents	1,600	400
Share of baseline	80.00%	20.00%
Female beneficiaries	671 (41.94%)	160 (40.00%)
Youth beneficiaries	477 (29.81%)	126 (31.50%)
Average annual income	NGN 1,626,526.36	NGN 981,668.74
Economic resilience	30.81%	29.75%

The results show broadly similar resilience levels between farmers and MSMEs, although farmers have a slightly higher baseline resilience rate.

The relatively similar female participation rates across both groups indicate that gender inclusion should be monitored across the full beneficiary portfolio rather than only within the farmer population.

9.2 Geographic Disaggregation

The baseline population was distributed across seven project LGAs.

LGA	Respondents	Share
Chikun	350	17.50%
Kaduna North	300	15.00%
Kaduna South	300	15.00%
Igabi	300	15.00%
Kajuru	250	12.50%
Kachia	250	12.50%
Zaria	250	12.50%

Geographic disaggregation establishes the baseline population required for future LGA-level performance monitoring.

The beneficiary composition also varies geographically. Chikun, Igabi, Kaduna North, Kaduna South, and Kajuru are represented primarily by farmers, while Kachia includes both farmers and MSMEs and Zaria is represented by MSME respondents in the synthetic baseline.

This geographic structure should be retained in future monitoring datasets to allow management to identify implementation differences between locations.

9.3 Agricultural Value-Chain Performance

The farmer baseline shows substantial differences in productivity and annual income across value chains.

Value Chain	Farmers	Average Productivity (t/ha)	Average Annual Income (NGN)
Tomato	155	5.42	3,619,051
Vegetables	187	4.16	2,500,886
Ginger	139	2.07	2,412,602
Rice	296	1.97	1,244,835
Soybean	219	1.90	1,131,995
Sorghum	135	1.50	1,071,685
Maize	386	1.80	1,039,586
Groundnut	83	1.37	876,347

Tomato records the strongest baseline productivity at **5.42 tonnes per hectare**, followed by vegetables at **4.16 tonnes per hectare**.

Groundnut records the lowest productivity at **1.37 tonnes per hectare**.

Tomato also records the highest average annual income at approximately **NGN 3.62 million**, while groundnut records the lowest at approximately **NGN 876,347**.

These differences indicate that value-chain-specific strategies may be more appropriate than applying identical productivity and income interventions across all farmer groups.

9.4 Agricultural Practice Adoption

Adoption of individual recommended practices varies considerably.

Practice	Farmers Adopting	Adoption Rate
Fertilizer management	857	53.56%
Improved seed	727	45.44%
Water management	677	42.31%
Improved storage	618	38.63%
Integrated pest management	614	38.38%
Climate-smart practice	553	34.56%
Conservation agriculture	544	34.00%
Soil testing	433	27.06%

Fertilizer management has the highest adoption rate at **53.56%**, while soil testing has the lowest at **27.06%**.

At the composite level, **67.81% of farmers adopt at least three recommended practices**.

This indicates that the project starts from a relatively strong general practice-adoption position but may need more targeted support for specific practices with lower adoption, particularly soil testing, conservation agriculture, and climate-smart practices.

9.5 Market Access

Market access indicators show important gaps between general buyer relationships and structured market participation.

Market Indicator	Farmers	Rate
Farmers with regular buyers	883	55.19%
Farmers selling through structured markets	437	27.31%
Farmers receiving market information	751	46.94%
Farmers reporting improved buyer access	480	30.00%

More than half of farmers report having regular buyers, but only **27.31% sell through structured markets**.

This suggests that having a regular buyer does not necessarily translate into participation in more formal or structured market arrangements.

The gap between market information access and improved buyer access also indicates that information provision alone may not be sufficient. Market interventions may need to combine information with aggregation, buyer engagement, quality standards, logistics, and transaction facilitation.

9.6 Post-Harvest Loss by Value Chain

The official OUT-06 post-harvest loss definition was used consistently for value-chain comparisons.

Value Chain	Post-Harvest Loss Rate
Vegetables	17.86%
Sorghum	17.71%
Tomato	17.49%
Soybean	17.34%
Rice	17.32%
Maize	17.21%
Ginger	17.10%
Groundnut	16.17%
Overall	**17.33%**

Vegetables record the highest average post-harvest loss rate at **17.86%**, while groundnut records the lowest at **16.17%**.

The relatively narrow range across value chains indicates that post-harvest loss is a cross-cutting issue rather than one confined to a single commodity.

9.7 Agricultural MSME Performance

The baseline profile for the 400 agricultural MSMEs is:

Measure	Baseline
MSMEs assessed	400
Average years in operation	10.43 years
Average employees	15.84
Average annual sales	NGN 988,565
Average annual operating cost	NGN 631,726
Average annual profit	NGN 356,839
Average profit margin	27.72%
MSMEs maintaining basic financial records	50.00%

The MSME results indicate an established business population, with an average of approximately 10.4 years in operation.

However, financial-management practices remain uneven. Only half of assessed MSMEs maintain at least three basic financial record types.

9.8 MSME Financial Record Keeping

The distribution of financial record types is:

Number of Record Types	MSMEs	Share
0	14	3.50%
1	71	17.75%
2	115	28.75%
3	136	34.00%
4	57	14.25%
5	7	1.75%

The most commonly maintained record types are:

- Sales records: 60.00%
- Expense records: 58.75%
- Inventory records: 51.00%
- Cash-flow records: 40.00%
- Profit and loss records: 33.25%

Profit-and-loss and cash-flow records are particularly weak areas and may require targeted financial-management support.

9.9 Gender and Youth

Women represent **41.55% of direct beneficiaries**, while youth aged 18-35 represent **30.17% of respondents with valid age information**.

Among farmers:

- Women represent **41.94%**.
- Youth represent **29.81%**.

Among MSMEs:

- Women represent **40.00%**.
- Youth represent **31.50%**.

Women's economic outcomes are substantially weaker than their participation level. Only **29.12% of women beneficiaries report increased agricultural or business income**.

This indicates that monitoring should distinguish between participation and actual economic benefit.

9.10 Economic Resilience

Overall, **30.60% of beneficiaries** are classified as economically resilient at baseline.

Group	Resilience Rate
Farmers	30.81%
MSMEs	29.75%
Female beneficiaries	29.96%
Male beneficiaries	31.05%
Youth	32.01%
Non-youth	29.94%

Resilience levels are broadly similar across the main beneficiary categories.

Youth beneficiaries show a slightly higher baseline resilience rate than non-youth beneficiaries, while female beneficiaries have a slightly lower rate than male beneficiaries.

These differences should be monitored over time to determine whether project interventions are producing equitable resilience gains.

9.11 Management Interpretation of Disaggregated Results

The disaggregated analysis demonstrates that aggregate project results can conceal important differences between beneficiary groups and value chains.

Key implications include:

1. **Value-chain differentiation:** Tomato and vegetables demonstrate substantially higher productivity and income than several other value chains, while groundnut requires particular attention.
2. **Practice-specific support:** Overall adoption is strong, but individual practices such as soil testing remain relatively weak.
3. **Market-system strengthening:** Regular buyer relationships are more common than structured market participation, indicating a need for deeper market-system interventions.
4. **Cross-cutting post-harvest challenge:** Loss rates are relatively high across all value chains.
5. **MSME financial management:** Record keeping remains an important constraint, particularly for cash-flow and profit-and-loss records.
6. **Gender outcome focus:** Women's participation is relatively strong, but women's economic outcomes remain weak.
7. **Resilience:** Economic resilience remains substantially below the 2030 target across both farmers and MSMEs.

Disaggregated monitoring should therefore remain an integral part of the ALMDP M&E system throughout implementation.

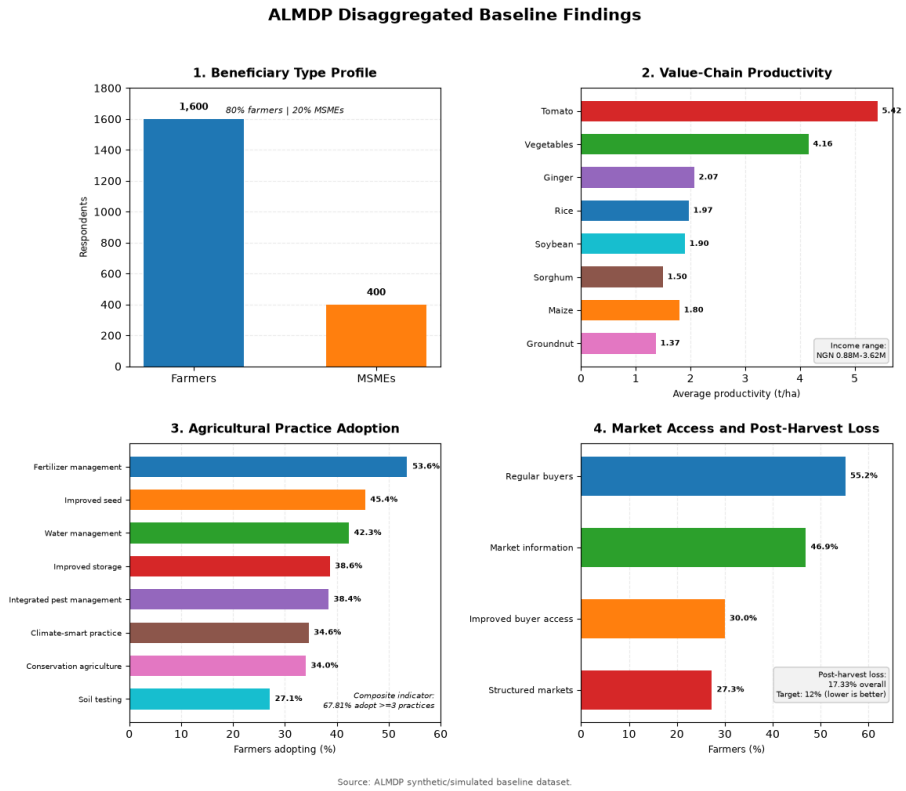


Figure 9. Disaggregated baseline findings across beneficiary groups, value chains, practices, and market access.

10. Indicator Gap Analysis

10.1 Overview

Indicator gap analysis compares validated baseline values with the approved 2030 targets in the ALMDP results framework.

The purpose is not to judge implementation performance at baseline. Instead, it establishes the distance between the starting position and the desired end-of-project result and helps management identify areas requiring early attention.

For higher-is-better indicators, the gap is calculated as:

Gap = 2030 Target - Baseline

For lower-is-better indicators, such as OUT-06, the gap is interpreted in the opposite direction:

Gap = Baseline - 2030 Target

Indicators that cannot yet be measured are retained as **Not measurable** rather than assigned artificial values.

10.2 Priority Classification

For management purposes, measurable outcome and impact indicators are classified according to the size and programme significance of their baseline gap.

The classification used in this assessment is:

Classification	Interpretation
Target met/exceeded	Baseline already meets or exceeds the 2030 target
Lower gap	Baseline is relatively close to the target
Moderate gap	Baseline requires meaningful improvement
High gap	Baseline is substantially below the target
Not measurable	Baseline cannot yet be calculated

The classification supports prioritisation but does not replace management judgement.

10.3 Impact and Outcome Gap Analysis

Indicator	Baseline	2030 Target	Gap	Direction	Priority
IMP-01: Average annual beneficiary income	NGN 1,497,879.70	NGN 600,000	Target exceeded	Higher is better	Target met/exceeded
IMP-02: Economic resilience	30.60%	65.00%	34.40 pp	Higher is better	High
OUT-01: Crop productivity	2.448 t/ha	2.500 t/ha	0.052 t/ha	Higher is better	Lower
OUT-02: Adoption of ≥3 practices	67.81%	65.00%	Target exceeded	Higher is better	Target met/exceeded
OUT-03: Adequate knowledge among trained farmers	Not measurable	75.00%	Not measurable	Higher is better	Not measurable
OUT-04: Structured market participation	27.31%	55.00%	27.69 pp	Higher is better	High
OUT-05: Improved buyer access	30.00%	70.00%	40.00 pp	Higher is better	High
OUT-06: Post-harvest loss rate	17.33%	12.00%	5.33 pp	Lower is better	Moderate
OUT-07: MSMEs reporting increased revenue	Not measurable	60.00%	Not measurable	Higher is better	Not measurable
OUT-08: Average MSME annual profit	NGN 356,839.05	NGN 550,000	NGN 193,160.95	Higher is better	Moderate
OUT-09: MSMEs maintaining basic records	50.00%	70.00%	20.00 pp	Higher is better	Moderate
OUT-10: Women beneficiaries	41.55%	45.00%	3.45 pp	Higher is better	Lower
OUT-11: Youth beneficiaries	30.17%	35.00%	4.83 pp	Higher is better	Lower
OUT-12: Women reporting increased income	29.12%	60.00%	30.88 pp	Higher is better	High

10.4 Highest-Priority Outcome Gaps

Four measurable outcome indicators represent the most substantial baseline gaps:

1. Improved Buyer Access - OUT-05

Baseline performance is **30.00%** against a target of **70.00%**, producing a **40 percentage-point gap**.

This is the largest measurable outcome gap and should be treated as a major market-system priority.

2. Economic Resilience - IMP-02

Baseline resilience is **30.60%** against a target of **65.00%**, producing a **34.40 percentage-point gap**.

The result indicates that many beneficiaries may remain vulnerable to agricultural, market, financial, or other economic shocks.

3. Women's Economic Outcomes - OUT-12

Only **29.12%** of women beneficiaries report increased agricultural or business income at baseline, compared with a target of **60.00%**.

The **30.88 percentage-point gap** indicates that beneficiary participation should not be treated as sufficient evidence of gender-responsive results.

4. Structured Market Participation - OUT-04

Only **27.31%** of farmers sell through structured markets, compared with a target of **55.00%**.

The **27.69 percentage-point gap** indicates a substantial need for market-linkage and value-chain development activities.

10.5 Moderate Gaps

Three indicators require meaningful but comparatively smaller improvements:

- **OUT-08:** MSME annual profit is NGN 356,839.05 against NGN 550,000, a gap of NGN 193,160.95.
- **OUT-09:** Financial record keeping is 50.00% against 70.00%, a 20 percentage-point gap.
- **OUT-06:** Post-harvest loss is 17.33% against a desired maximum of 12.00%, requiring a 5.33 percentage-point reduction.

These indicators should be incorporated into implementation workplans and monitored regularly.

10.6 Lower-Gap Indicators

Several indicators are already relatively close to their 2030 targets:

- **OUT-01:** Productivity is 2.448 t/ha against 2.50 t/ha.
- **OUT-10:** Women represent 41.55% against a target of 45%.
- **OUT-11:** Youth represent 30.17% against a target of 35%.

These indicators should still be monitored because being close to target at baseline does not guarantee that the target will be maintained or achieved.

10.7 Indicators Already Meeting or Exceeding the Target

Two indicators exceed their approved 2030 targets at baseline:

- **IMP-01:** Average annual beneficiary income is NGN 1,497,879.70 against a target of NGN 600,000.
- **OUT-02:** Adoption of at least three recommended agricultural practices is 67.81% against a target of 65%.

These results should not be interpreted as evidence of project success because implementation has not yet begun.

Instead, they establish strong starting positions and should prompt management to examine whether:

- the targets remain appropriate;
- measurement definitions remain consistent;
- the baseline values are sufficiently robust;
- future implementation can sustain or improve the starting position.

10.8 Indicators Not Yet Measurable

Five indicators cannot yet be assessed from the baseline dataset:

Indicator	Reason
OUT-03	Requires implementation-related training and knowledge assessment
OUT-07	Requires implementation-period revenue-change measurement
OP-08	Requires operational M&E activity records
OP-09	Requires implementation-period reporting records
OP-10	Requires implementation-period DQA records

These indicators should remain explicitly classified as **Not measurable** until the required data become available.

Assigning zero to these indicators would create misleading baseline results.

10.9 Management Priority Matrix

Priority	Indicators	Management Implication
High	IMP-02, OUT-04, OUT-05, OUT-12	Require early implementation attention and close monitoring
Moderate	OUT-06, OUT-08, OUT-09	Require targeted interventions and regular performance review
Lower	OUT-01, OUT-10, OUT-11	Maintain performance and monitor progress
Target met/exceeded	IMP-01, OUT-02	Protect gains and validate measurement assumptions
Not measurable	OUT-03, OUT-07, OP-08, OP-09, OP-10	Establish measurement systems before reporting performance

10.10 Overall Gap Analysis Conclusion

The baseline demonstrates a mixed starting position.

The strongest areas are beneficiary income, agricultural practice adoption, and proximity to the productivity, gender, and youth targets.

The most significant gaps concern:

- buyer access;
- economic resilience;
- women's economic outcomes;
- structured market participation;
- MSME profitability;
- financial record keeping; and
- post-harvest losses.

The gap analysis therefore supports an implementation strategy that prioritises **market-system strengthening, resilience, women's economic empowerment, post-harvest management, and MSME business development**, while maintaining the relatively stronger baseline position in agricultural practice adoption and productivity.

The identified gaps should be revisited during routine monitoring to determine whether implementation is closing the distance to the 2030 targets.

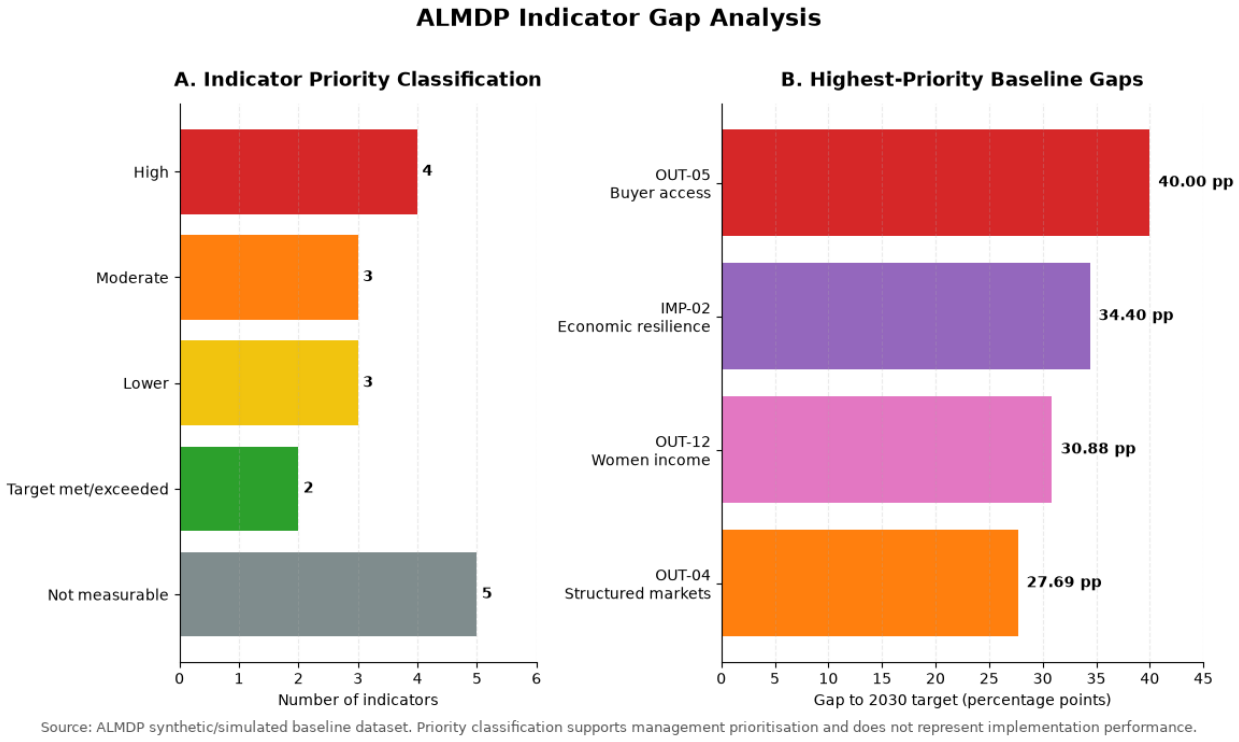


Figure 10. Indicator priority classification and highest-priority baseline gaps.

11. Baseline Findings and Management Implications

11.1 Overview

The baseline analysis identified 18 principal findings across programme performance, agricultural productivity, market access, MSME development, inclusion, resilience, data quality, and M&E systems.

The findings are intended to support evidence-based management decisions during the early stages of ALMDP implementation.

The baseline should be interpreted as a starting position rather than an assessment of project performance, because implementation has not yet begun and the dataset is synthetic.

11.2 Finding 1 - Overall Performance Position Is Mixed

The baseline presents a mixed performance profile.

Two indicators already meet or exceed their 2030 targets, several indicators are relatively close to target, while a number of outcomes show substantial gaps.

Management implication

Implementation planning should distinguish between areas requiring major improvement and areas where the priority is to maintain or consolidate a relatively strong starting position.

11.3 Finding 2 - Economic Resilience Is a Major Programme Gap

Only **30.60% of beneficiaries** meet the baseline resilience criterion compared with a target of **65%**.

The gap of **34.40 percentage points** makes resilience one of the most significant programme-level priorities.

Management implication

Implementation should integrate resilience-building measures such as:

- diversified income opportunities;
- improved market access;
- climate-smart production;
- savings and financial-management practices;
- improved business continuity;
- access to relevant services and support networks.

Resilience should be tracked across farmers, MSMEs, women, and youth rather than only at aggregate level.

11.4 Finding 3 - Market Participation Is Weak

Only **27.31% of farmers** report selling through structured markets, compared with a target of **55%**.

Although **55.19%** report having regular buyers, structured market participation remains considerably lower.

Management implication

Market interventions should go beyond general buyer identification and information dissemination.

Priority actions should include:

- aggregation mechanisms;
- buyer-seller linkage events;
- market information systems;
- quality and grading support;
- transaction facilitation;
- logistics coordination;
- stronger relationships with structured buyers.

11.5 Finding 4 - Buyer Access Is the Largest Measurable Outcome Gap

Only **30.00% of farmers** report improved buyer access against a target of **70%**.

This represents a **40 percentage-point gap**, the largest measurable gap among the assessed impact and outcome indicators.

Management implication

Buyer-access interventions should receive early implementation attention.

The programme should monitor not only whether beneficiaries receive market information but whether information results in actual buyer relationships, increased sales opportunities, improved prices, and repeat market participation.

11.6 Finding 5 - Agricultural Productivity Is Relatively Strong

Average crop productivity is **2.448 tonnes per hectare**, compared with a 2030 target of **2.50 tonnes per hectare**.

The baseline is therefore only approximately **0.052 tonnes per hectare below the target**.

Management implication

The programme should focus on sustaining and improving productivity rather than treating productivity as a major crisis area.

Targeted technical assistance should focus on value chains and locations with weaker productivity.

11.7 Finding 6 - Agricultural Practice Adoption Is Already Above Target

67.81% of farmers report adopting at least three recommended practices, compared with a target of **65%**.

This indicates a strong baseline position.

However, individual practice adoption varies considerably.

Fertilizer management has the highest adoption rate at **53.56%**, while soil testing has the lowest at **27.06%**.

Management implication

The programme should maintain the overall adoption level while targeting specific practices with lower uptake, particularly:

- soil testing;
- conservation agriculture;
- climate-smart practices;
- integrated pest management.

11.8 Finding 7 - Post-Harvest Loss Remains Above the Desired Level

The official baseline post-harvest loss rate is **17.33%**, compared with a desired target of **12%**.

The baseline therefore requires a reduction of approximately **5.33 percentage points**.

Loss rates remain relatively high across all assessed value chains.

Management implication

Post-harvest interventions should be treated as cross-cutting rather than commodity-specific.

Potential measures include:

- improved storage;
- improved handling;
- aggregation;
- transport management;
- packaging;
- market timing;
- post-harvest technology;
- farmer and MSME training.

11.9 Finding 8 - MSME Profitability Requires Improvement

Average MSME annual profit is **NGN 356,839.05**, compared with a target of **NGN 550,000**.

The baseline shortfall is approximately **NGN 193,160.95**.

Management implication

MSME support should address the drivers of profitability rather than focusing only on training participation.

Priority areas include:

- cost management;
- pricing;
- inventory management;
- market development;
- business planning;
- financial record keeping;
- cash-flow management.

11.10 Finding 9 - MSME Financial Record Keeping Is Incomplete

Only **50% of assessed MSMEs** maintain at least three basic financial record types.

Cash-flow and profit-and-loss records are particularly weak.

Management implication

Financial-management support should include practical systems that MSMEs can maintain consistently, rather than relying solely on classroom-based training.

Progress should be measured through actual records maintained, not only training attendance.

11.11 Finding 10 - Women's Participation Is Relatively Strong

Women represent **41.55% of direct beneficiaries**, compared with a target of **45%**.

The baseline gap is only **3.45 percentage points**.

Management implication

The programme should maintain women's participation while ensuring that inclusion translates into measurable economic benefits.

Participation targets should therefore be monitored alongside women's income, resilience, market access, and business performance.

11.12 Finding 11 - Women's Economic Outcomes Remain Weak

Only **29.12% of women beneficiaries** report increased agricultural or business income compared with a target of **60%**.

This represents a **30.88 percentage-point gap**.

Management implication

Gender-responsive implementation should focus on economic outcomes rather than participation alone.

Women may require targeted support related to:

- access to markets;
- productive assets;
- finance;
- business development;
- technology;
- training;
- networks and buyer relationships.

11.13 Finding 12 - Youth Participation Is Relatively Close to Target

Youth aged 18-35 represent **30.17%** of respondents with valid age information, compared with a target of **35%**.

The gap is approximately **4.83 percentage points**.

Management implication

Youth inclusion should be maintained and strengthened through targeted participation opportunities in production, aggregation, processing, trading, digital market services, and agricultural MSMEs.

11.14 Finding 13 - Value-Chain Performance Varies Substantially

The baseline shows major differences across agricultural value chains.

Tomato has the highest average productivity at **5.42 t/ha** and the highest average annual income at approximately **NGN 3.62 million**.

Groundnut has the lowest productivity at **1.37 t/ha** and the lowest average annual income at approximately **NGN 876,347**.

Management implication

Value-chain interventions should be differentiated.

The programme should identify:

- high-performing value chains that can provide lessons for other groups;
- lower-performing value chains requiring additional technical support;
- market opportunities;
- post-harvest constraints;

- value-chain-specific production risks.

11.15 Finding 14 - Post-Harvest Loss Is a Cross-Cutting Issue

Post-harvest loss rates range from approximately **16.17% to 17.86%** across the assessed value chains.

The relatively narrow range indicates that post-harvest loss is not isolated to one commodity.

Management implication

The programme should incorporate post-harvest management into multiple components, including farmer training, market access, storage, logistics, and MSME development.

11.16 Finding 15 - Several Indicators Are Not Yet Measurable

Five indicators cannot be measured from the baseline dataset:

- OUT-03: Adequate knowledge among trained farmers;
- OUT-07: MSMEs reporting increased annual revenue;
- OP-08: M&E activities completed on time;
- OP-09: Required reports submitted on time;
- OP-10: Records passing DQA.

Management implication

The project should establish the required measurement systems before implementation data are generated.

This includes:

- training attendance records;
- knowledge assessment tools;
- MSME financial monitoring;
- M&E activity trackers;
- reporting calendars;
- DQA records.

11.17 Finding 16 - Data Quality Controls Should Remain Active

The synthetic baseline dataset required several validation and cleaning actions, including ID collision resolution, invalid age handling, farm-area validation, production logic checks, and invalid knowledge-response correction.

After cleaning, the major validation checks passed.

Management implication

The same data-quality controls should be embedded into routine implementation monitoring.

Automated validation should be supplemented by manual review and periodic source verification.

11.18 Finding 17 - Disaggregation Is Essential for Management

Aggregate results conceal differences between:

- farmers and MSMEs;
- LGAs;
- agricultural value chains;
- women and men;
- youth and non-youth.

The baseline demonstrates that performance differs substantially across these groups.

Management implication

Routine monitoring should preserve the same disaggregation dimensions wherever appropriate.

Management dashboards should allow users to identify underperforming groups and locations rather than relying only on project-wide averages.

11.19 Finding 18 - Results-Based Management Should Drive Implementation

The baseline provides the reference point for a continuous management cycle:

Baseline evidence -> implementation -> monitoring -> variance analysis -> corrective action -> follow-up

The purpose of the M&E system is therefore not only to produce reports but to support timely management decisions.

Management implication

Programme teams should establish regular performance-review meetings where indicator trends, implementation bottlenecks, data-quality issues, and corrective actions are documented and followed up.

11.20 Consolidated Management Priorities

Based on the findings, the following priorities should receive particular attention during early implementation:

Priority	Management Focus	Key Indicators
1	Strengthen market access and buyer linkages	OUT-04, OUT-05
2	Improve economic resilience	IMP-02
3	Strengthen women's economic outcomes	OUT-12
4	Reduce post-harvest losses	OUT-06
5	Improve MSME profitability and financial management	OUT-08, OUT-09
6	Sustain agricultural productivity and practice adoption	OUT-01, OUT-02
7	Maintain gender and youth participation	OUT-10, OUT-11
8	Establish systems for currently unmeasurable indicators	OUT-03, OUT-07, OP-08, OP-09, OP-10
9	Maintain strong data-quality controls	All indicators
10	Institutionalise results-based management	Full results framework

11.21 Conclusion

The baseline provides a structured evidence base for ALMDP implementation planning and future performance monitoring.

The programme begins with relatively strong agricultural practice adoption and crop productivity, but substantial gaps remain in market participation, buyer access, economic resilience, women's economic outcomes, post-harvest management, and MSME financial performance.

The most important management principle emerging from the baseline is that **participation alone should not be treated as evidence of improved livelihoods**. Future monitoring should examine whether programme activities translate into measurable improvements in productivity, income, market access, business performance, resilience, and equitable economic outcomes.

These findings provide the basis for subsequent target monitoring, indicator tracking, management action, and annual performance assessment.

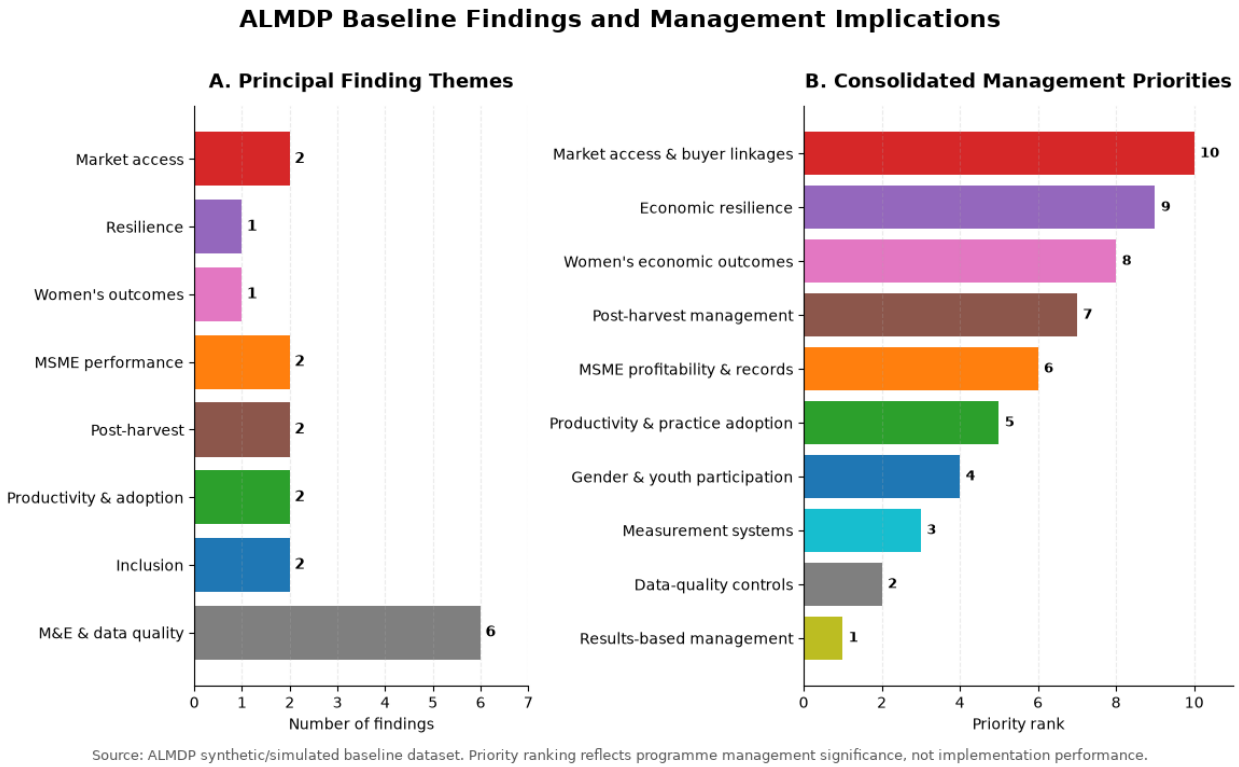


Figure 11. Principal finding themes and consolidated management priorities.

12. Baseline Conclusions and Recommendations

12.1 Baseline Conclusion

The ALMDP synthetic baseline assessment establishes a structured starting position for monitoring implementation progress across agricultural livelihoods, agricultural MSMEs, market participation, inclusion, resilience, and M&E system performance.

The assessment covered **2,000 synthetic respondents**, comprising **1,600 farmers and 400 agricultural MSMEs**, across seven project LGAs and eight agricultural value chains.

The baseline demonstrates a mixed starting position.

Agricultural practice adoption is relatively strong, with **67.81% of farmers** reporting adoption of at least three recommended practices, exceeding the 2030 target of 65%.

Average crop productivity is also relatively close to the target, at **2.448 tonnes per hectare** compared with 2.50 tonnes per hectare.

Average annual beneficiary income is above the approved 2030 target in the synthetic baseline. This result should be treated as a starting reference value rather than evidence of project achievement because the project is pre-implementation.

At the same time, several important development gaps are evident.

Economic resilience is only **30.60%**, compared with a target of 65%. Structured market participation is **27.31%**, compared with 55%, while improved buyer access is **30.00%**, compared with 70%.

Women's economic outcomes are also weak. Only **29.12% of women beneficiaries** report increased agricultural or business income, compared with a target of 60%.

Post-harvest losses remain above the desired level at **17.33%**, compared with a target of 12%.

Agricultural MSME profitability also requires improvement. Average annual profit is approximately **NGN 356,839**, compared with a target of NGN 550,000, while only **50% of assessed MSMEs** maintain at least three basic financial record types.

The baseline therefore suggests that ALMDP implementation should prioritise **market access, economic resilience, women's economic empowerment, post-harvest management, and MSME business performance**, while sustaining the relatively strong starting position in agricultural practice adoption and productivity.

12.2 Key Conclusions

Conclusion 1 - The programme has a strong basis for agricultural practice adoption

The proportion of farmers adopting at least three recommended practices already exceeds the 2030 target.

The implementation challenge is therefore less about creating basic awareness and more about sustaining adoption, improving the quality of adoption, and strengthening lower-performing practices.

Conclusion 2 - Market-system constraints require substantial attention

The difference between regular buyer relationships and structured market participation demonstrates that farmers may have market contacts without having reliable access to formal or structured market opportunities.

Market interventions should therefore focus on the complete pathway from production to sale.

Conclusion 3 - Economic resilience requires a multi-dimensional response

Low resilience levels across both farmers and MSMEs suggest that resilience should not be addressed through a single intervention.

Production, market access, financial management, income diversification, climate adaptation, and business continuity should be considered together.

Conclusion 4 - Women's participation must translate into economic benefit

Women's representation among beneficiaries is relatively close to the project target.

However, the substantially lower level of reported women's economic gains demonstrates that participation indicators alone are insufficient.

The M&E system should therefore track both inclusion and outcome-level economic benefits.

Conclusion 5 - MSME development should focus on business performance

Training participation should not be treated as the primary measure of MSME success.

The programme should monitor whether business support leads to improved financial management, profitability, revenue growth, market access, and resilience.

Conclusion 6 - Post-harvest management is a cross-cutting issue

The relatively similar post-harvest loss rates across value chains suggest that post-harvest management should be integrated into multiple programme components.

Conclusion 7 - Disaggregation should remain central to M&E

The baseline demonstrates meaningful differences between beneficiary groups and value chains.

Future monitoring should preserve disaggregation by:

- beneficiary type;
- LGA;
- value chain;
- sex;
- age group;
- and other approved programme characteristics.

Conclusion 8 - Data systems should be established before implementation reporting

Five indicators are not measurable from the baseline dataset because they require implementation-period information.

The necessary systems should therefore be established before the first routine performance-reporting cycle.

12.3 Recommendations

Recommendation 1 - Strengthen market access and buyer linkages

Priority: Very High

The programme should establish a coordinated market-linkage strategy addressing:

- aggregation;
- structured buyer relationships;
- market information;
- quality standards;
- grading;
- transaction facilitation;
- logistics;
- buyer-seller engagement.

Relevant indicators: OUT-04 and OUT-05.

Recommendation 2 - Integrate economic resilience across programme components

Priority: Very High

Resilience should be integrated into agricultural productivity, market access, MSME development, climate-smart agriculture, and financial-management interventions.

Monitoring should track resilience by beneficiary type, sex, age group, LGA, and other approved disaggregation dimensions.

Relevant indicator: IMP-02.

Recommendation 3 - Strengthen women's economic empowerment

Priority: Very High

The programme should move beyond participation targets and monitor whether women experience measurable improvements in:

- agricultural income;
- business income;
- market access;

- productive assets;
- financial services;
- business performance;
- resilience.

Relevant indicator: OUT-12, supported by OUT-10.

Recommendation 4 - Implement a cross-cutting post-harvest-loss strategy

Priority: High

Interventions should address:

- storage;
- handling;
- packaging;
- transport;
- aggregation;
- processing;
- market timing;
- post-harvest technology.

The programme should monitor losses by value chain and location.

Relevant indicator: OUT-06.

Recommendation 5 - Strengthen MSME financial management and profitability

Priority: High

Business-support interventions should include practical tools for:

- sales recording;
- expense recording;
- inventory management;
- cash-flow management;
- profit-and-loss reporting;
- costing;
- pricing;
- business planning.

Support should be linked to actual business performance rather than training attendance alone.

Relevant indicators: OUT-08 and OUT-09.

Recommendation 6 - Sustain agricultural productivity and practice adoption

Priority: Moderate

Because the baseline is already close to or above target for productivity and practice adoption, implementation should protect existing performance while addressing weaker practices.

Particular attention should be given to:

- soil testing;
- conservation agriculture;
- climate-smart practices;
- integrated pest management.

Relevant indicators: OUT-01 and OUT-02.

Recommendation 7 - Maintain gender and youth inclusion

Priority: Moderate

Women and youth participation should remain active implementation priorities.

However, inclusion should be connected to meaningful outcomes such as income, market access, resilience, and business performance.

Relevant indicators: OUT-10 and OUT-11.

Recommendation 8 - Establish systems for currently unmeasurable indicators

Priority: High

Before routine reporting begins, the programme should establish:

- training attendance and completion records;
- knowledge assessment tools;
- MSME revenue tracking;
- M&E activity calendars;
- reporting trackers;

- DQA records.

Relevant indicators: OUT-03, OUT-07, OP-08, OP-09, and OP-10.

Recommendation 9 - Institutionalise data-quality assurance

Priority: High

Routine monitoring should include:

- automated validation;
- duplicate checks;
- completeness checks;
- logical consistency checks;
- range checks;
- source-document verification;
- back-checks;
- spot checks;
- record-to-report reconciliation;
- periodic DQA assessments.

The DQA protocol should be applied consistently across data-collection cycles.

Recommendation 10 - Institutionalise results-based management

Priority: High

Programme management should use the M&E system as a decision-support mechanism rather than solely as a reporting function.

Performance reviews should follow the cycle:

Evidence -> variance -> investigation -> corrective action -> responsibility -> deadline -> follow-up

Management decisions should be documented and tracked to completion.

12.4 Recommended Management Action Matrix

Priority	Action	Responsible Function	Indicative Timing	Key Indicators
Very High	Develop market-linkage strategy	Market/Value-Chain Team	Early implementation	OUT-04, OUT-05
Very High	Integrate resilience interventions	Programme/Technical Team	Early implementation	IMP-02
Very High	Develop women's economic empowerment actions	Gender/Programme Team	Early implementation	OUT-10, OUT-12
High	Implement post-harvest-loss measures	Agriculture/Value-Chain Team	Early implementation	OUT-06
High	Strengthen MSME financial-management support	MSME Team	Early implementation	OUT-08, OUT-09
High	Activate currently unmeasurable indicators	M&E Team	Before routine reporting	OUT-03, OUT-07, OP-08-OP-10
High	Maintain routine DQA	M&E/Data Team	Every reporting cycle	OP-10
Moderate	Sustain agricultural productivity and adoption	Agriculture Team	Throughout implementation	OUT-01, OUT-02
Moderate	Maintain gender and youth inclusion	Programme/Gender Team	Throughout implementation	OUT-10, OUT-11
High	Institutionalise management review meetings	Project Management	Monthly/Quarterly	Full results framework

12.5 Recommended Monitoring Priorities for Early Implementation

The first implementation monitoring cycles should place particular attention on:

1. Market-linkage activities and buyer engagement.
2. Economic resilience interventions.
3. Women's economic participation and outcomes.
4. Post-harvest management.
5. MSME financial-management support.
6. Agricultural practice adoption by individual practice.
7. Productivity by value chain.
8. Youth and women's participation.
9. Data quality and completeness.
10. Activation of indicators currently classified as not measurable.

12.6 Final Conclusion

The baseline assessment provides a coherent quantitative reference point for ALMDP implementation monitoring.

The evidence suggests that the programme does not begin from a uniformly weak position. Several areas demonstrate relatively strong baseline conditions, particularly agricultural practice adoption and crop productivity.

However, substantial gaps remain in market participation, buyer access, economic resilience, women's economic outcomes, post-harvest management, and MSME financial performance.

The central M&E priority should therefore be to determine whether programme implementation converts activities and outputs into **measurable improvements in livelihoods, income, market access, resilience, business performance, and equitable economic outcomes**.

Because this is a synthetic portfolio assessment, the numerical results should be treated as simulated analytical evidence demonstrating the design and application of an M&E system rather than as actual Kaduna State programme results.

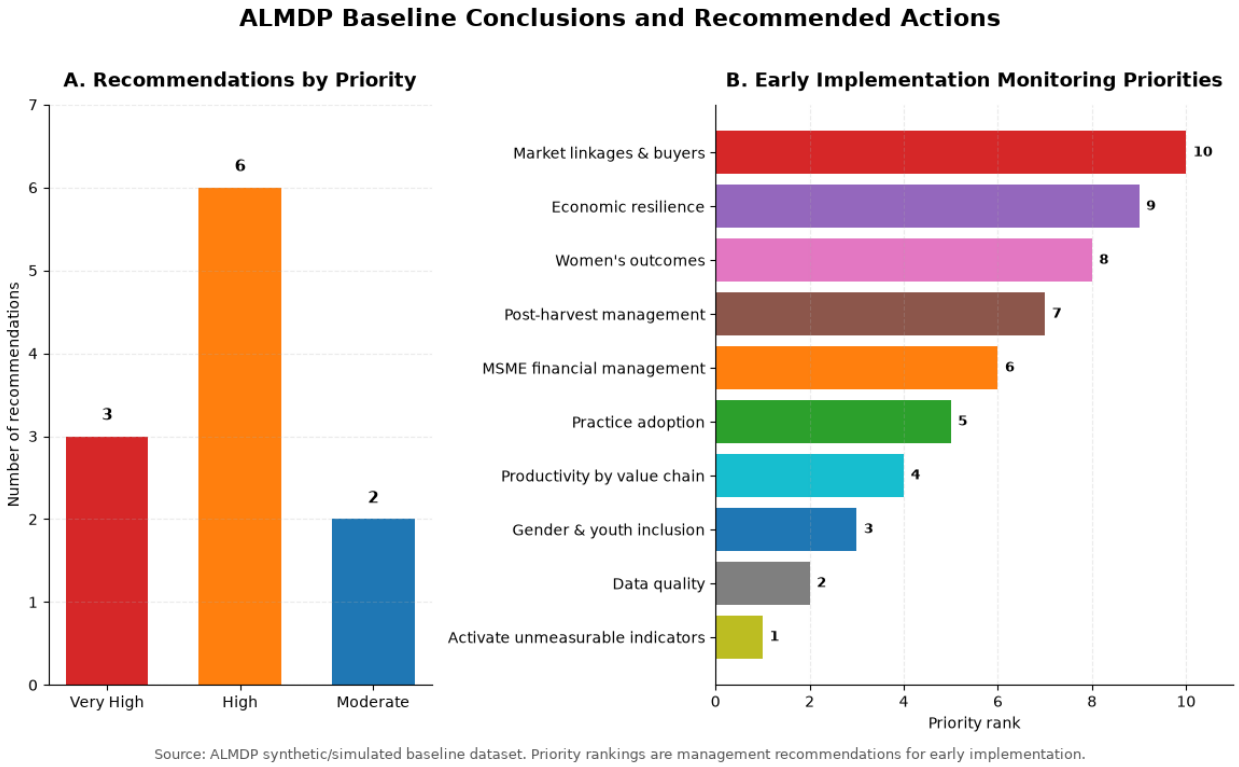


Figure 12. Recommendations by priority and early implementation monitoring priorities.